

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

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Forename(s)

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Candidate signature

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INTERNATIONAL GCSE BIOLOGY

Paper 1

Tuesday 6 November 2018 07:00 GMT Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a ruler with millimetre measurements
- a scientific calculator.

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- There are 90 marks available on this paper.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers.

Advice

In all calculations, show clearly how you work out your answer.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
TOTAL	



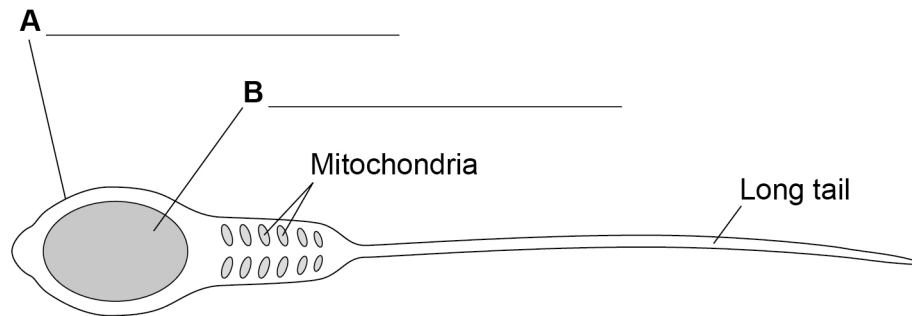
Answer **all** the questions in the spaces provided.

Do not write
outside the
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0 1

Figure 1 shows a human sperm cell.

Figure 1



0 1 . 1

Label parts **A** and **B** on **Figure 1**.

Use words from the box below.

[2 marks]

Cell membrane

Cell wall

Cytoplasm

Nucleus

Plasmid



0 1 . 2 In **Figure 1** the microscope image of the sperm cell has a length of 12.5 mm.

The sperm cell has been magnified 2500 times.

Calculate the actual length of the sperm cell in mm.

Use the equation:

$$\text{Actual length} = \frac{\text{image length}}{\text{magnification}}$$

[2 marks]

Actual length = _____ mm

0 1 . 3 Sperm cells contain many mitochondria.

Describe how the mitochondria help the sperm to do its job.

[2 marks]

Question 1 continues on the next page

Turn over ►

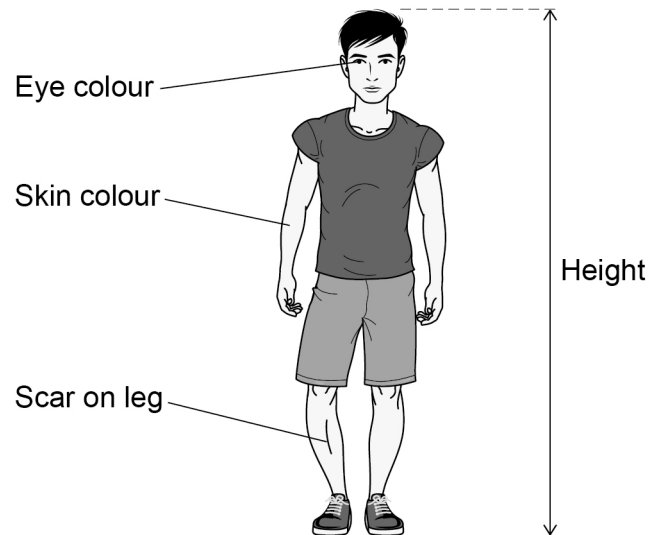


Figure 2 shows four characteristics of a man.

The characteristics may be due to:

- the environment **only**
- genes **only**
- both genes **and** the environment.

Figure 2



0 1 . 4

Complete **Table 1**.

Tick (✓) **one** box in each row to show what each characteristic is due to.

[2 marks]

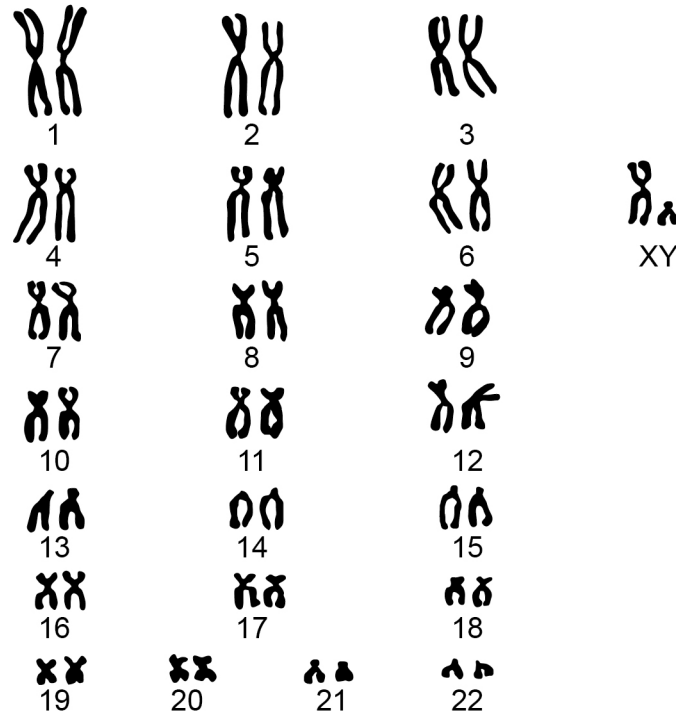
Table 1

Characteristic	Characteristic is due to		
	Environment only	Genes only	Genes and the environment
Eye colour			
Height			
Scar on leg			
Skin colour			



Figure 3 shows the full set of chromosomes from a human male.

Figure 3



0 1 . 5

How can you tell from **Figure 3** that this person is male?

[1 mark]

0 1 . 6

How many chromosomes are there in a human sperm cell?

[1 mark]

Question 1 continues on the next page

Turn over ►



0 1 . 7 A sperm cell and an egg cell join at fertilisation.

The fertilised egg cell then divides to form an embryo.

Describe how the fertilised egg divides to form an embryo.

[3 marks]

13



0 2

Vitamin A is an important part of the diet.

'Golden rice' has been genetically engineered to provide vitamin A.

0 2 . 1

Steps 1–3 show the stages in genetic engineering.

Complete the sentences.

Use words from the box.

[3 marks]

an early		an embryo		an enzyme		a herbicide			
a late		the middle		a vaccine		a vector		a ribosome	

Step 1: Scientists isolate the required gene using _____.

Step 2: This gene is transferred into cells of an organism using _____.

Step 3: Genes must be transferred to organisms when they are
at _____ stage of development.

Question 2 continues on the next page

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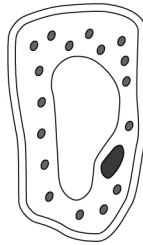


0 3

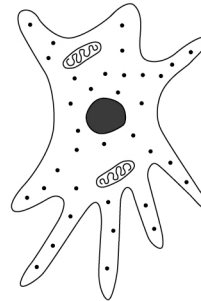
Figure 4 shows two different cells. Fibroblast cells can be found in animals.

Figure 4

Plant cell



Fibroblast cell



0 3 . 1

Give **two** structures found in plant cells that are **not** found in an animal cell.

[2 marks]

1 _____

2 _____

0 3 . 2

Fibroblast cells make protein fibres.

Which parts of a cell make proteins?

[1 mark]

Tick (✓) **one** box.

Cell membrane

☐

Mitochondria

☐

Ribosomes

☐

Question 3 continues on the next page

Turn over ►



0 3 . 3 Which structure in the fibroblast cell contains chromosomes?

[1 mark]

0 3 . 4 Chromosomes carry genes.

What is a gene?

[2 marks]



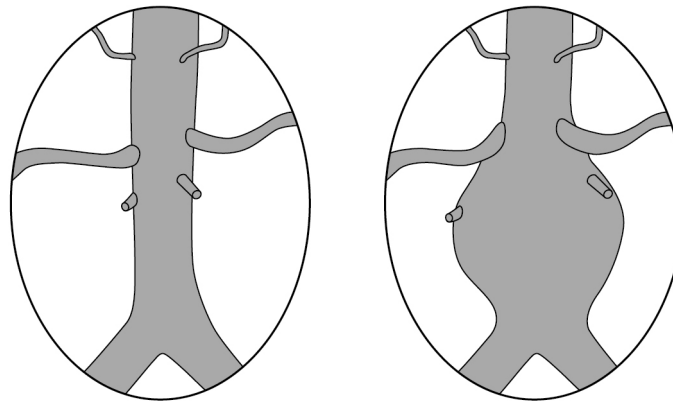
0 3 . 5 Ehlers Danlos syndrome (EDS) is a genetic disorder caused by a mutated gene.

The fibroblast cells in a person with EDS make very few protein fibres.

Protein fibres provide support for tissues.

Look at **Figure 5**.

Figure 5



Normal artery

EDS artery

Suggest why a person with EDS might bleed from an artery.

[2 marks]

Question 3 continues on the next page

Turn over ►

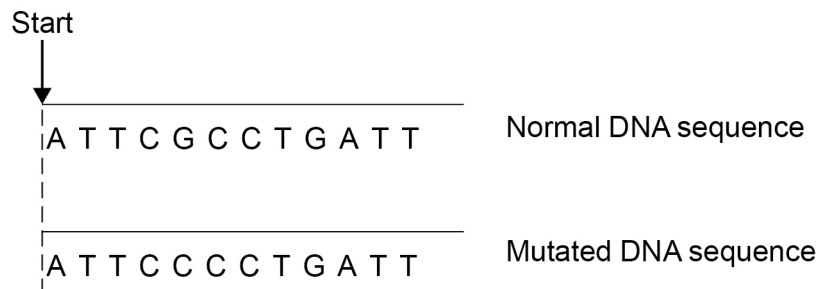


The DNA in the mutated EDS allele has a different sequence of bases from the DNA in a normal allele.

Figure 6 shows the sequence of bases in the same section of DNA from a normal allele and from a mutated EDS allele.

The bases are represented by the letters A, T, C and G.

Figure 6



0 3 . 6 Identify how the bases have changed in the mutated DNA sequence.

[1 mark]

0 3 . 7 **Table 2** shows DNA base sequences and the amino acids they code for.

Table 2

DNA base sequence	Amino acids
ATT, ATA	Isoleucine
CGA, CGT, CGC	Arginine
CTA, CTG, CTT	Leucine
CCC, CCG	Proline

How will the mutated DNA base sequence alter the amino acid order?

[1 mark]



0 3 . 8 EDS is caused by a dominant allele.

Two parents have a child who does **not** have EDS.

The mother has EDS.

The father does **not** have EDS.

Draw a genetic diagram to find the probability that the parents' next child will have EDS.

You should identify in your diagram any children who have EDS.

Use the following symbols: **D** = allele for EDS
d = allele for **not** having EDS

[4 marks]

Probability = _____

14

Turn over ►



0 4

This question is about receptors.

0 4 . 1

Name the organs where these receptors are found.

[2 marks]

1. Light receptors _____

2. Sound receptors _____

0 4 . 2

Receptors are involved in reflex actions.

Describe the nervous pathway taken during a pain withdrawal reflex.

[4 marks]



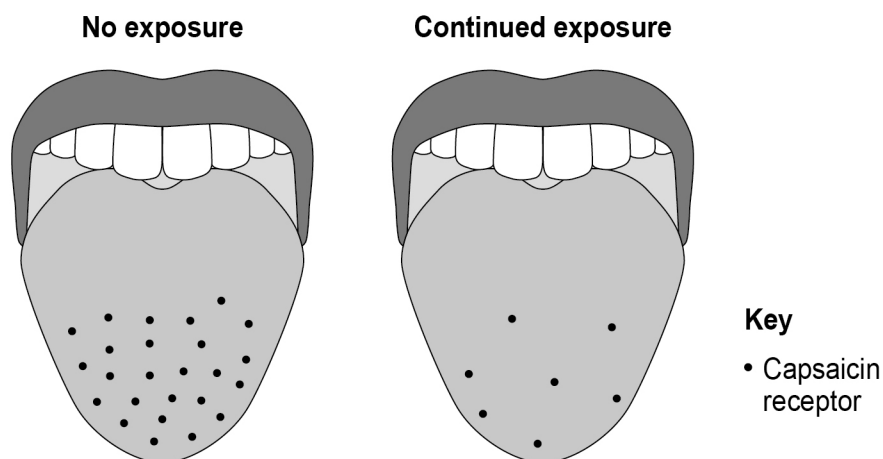
There are receptors in the tongue.

Capsaicin is a substance found in chilli peppers.

Capsaicin pain receptors on the tongue are stimulated when a person eats a chilli pepper.

Continued exposure to capsaicin causes a change to the receptors in the tongue as shown in **Figure 7**.

Figure 7



0 4 . 3

Explain why people who have eaten chilli peppers for many years are less likely to feel pain when eating chilli peppers.

[2 marks]

Question 4 continues on the next page

Turn over ►



Table 3 shows the solubility in water of capsaicin and some other chemicals at 40 °C.

Table 3

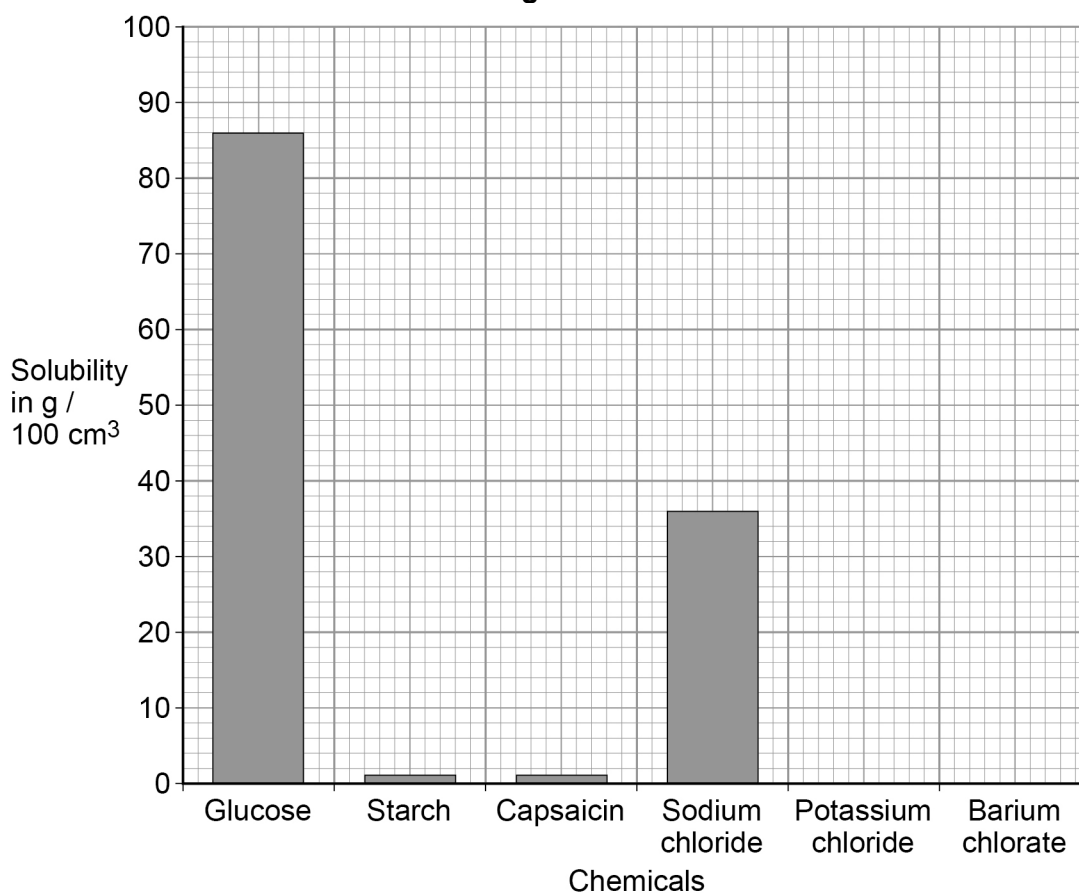
Chemicals	Solubility in g / 100cm ³
Glucose	86
Starch	1
Capsaicin	1
Sodium chloride	36
Potassium chloride	42
Barium chlorate	25

0 4 . 4 Complete the bar chart in **Figure 8**.

Use the data in **Table 3**.

[1 mark]

Figure 8



0 4 . 5

Suggest why drinking water does **not** remove the pain immediately after eating chilli peppers.

[2 marks]

0 4 . 6

A plant leaf cell makes glucose in photosynthesis. Most of the glucose will be immediately converted to starch and stored within the cell.

Explain why storing starch rather than glucose stops plant cells from swelling with too much water.

Use **Figure 8** and your own knowledge.

[3 marks]

14

Turn over for the next question

Turn over ►

0 5

Glucose is produced in photosynthesis.

0 5 . 1

Complete a **balanced symbol** equation for photosynthesis.**[2 marks]**

Plants and water snails can be used to investigate photosynthesis and respiration.

Hydrogencarbonate indicator changes colour in different concentrations of carbon dioxide as shown in **Table 4**.**Table 4**

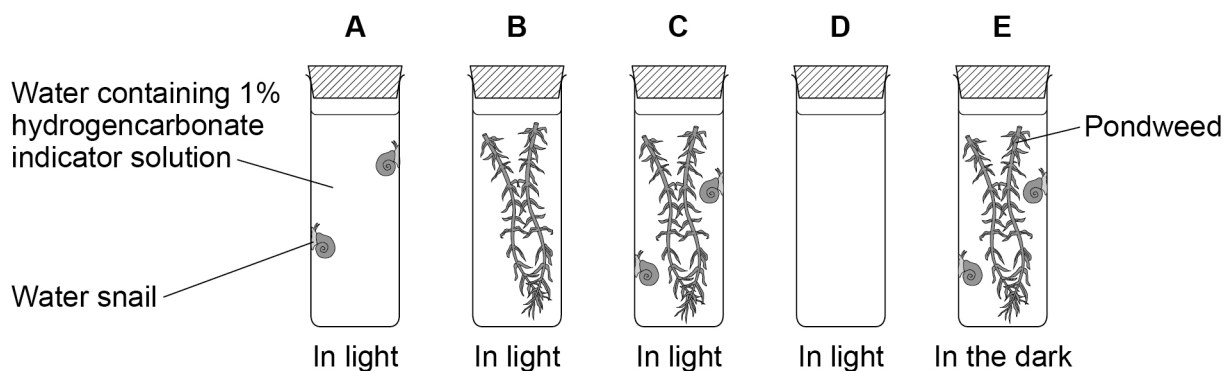
Concentration of carbon dioxide	Colour of indicator
Remains constant	Red
High	Yellow
Low	Purple

Hydrogencarbonate indicator does not affect plants or animals.

Some students investigated photosynthesis and respiration.

This is the method used.

1. Set up five plastic tubes as shown in **Figure 9**.
2. Leave tubes A, B, C and D in the light.
3. Put tube E in the dark.
4. Record the colour of the indicator in all tubes after 12 hours.

Figure 9

0 5 . 2

Give **two** variables the students should have controlled in their investigation.**[2 marks]**

1 _____

2 _____

At the start of the investigation the colour of the indicator solution in all tubes was red.

Table 5 shows the results.

Table 5

Tube	Colour after 12 hours
C, D	Red
A, E	Yellow
B	Purple

Question 5 continues on the next page

Turn over ►



[5 marks]

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

[2 marks]

Reason



0 5 . 5 Identify **one** hazard in the investigation.

State **one** safety precaution needed to reduce the risk of this hazard.

[2 marks]

Hazard

Safety precaution

13

Turn over for the next question

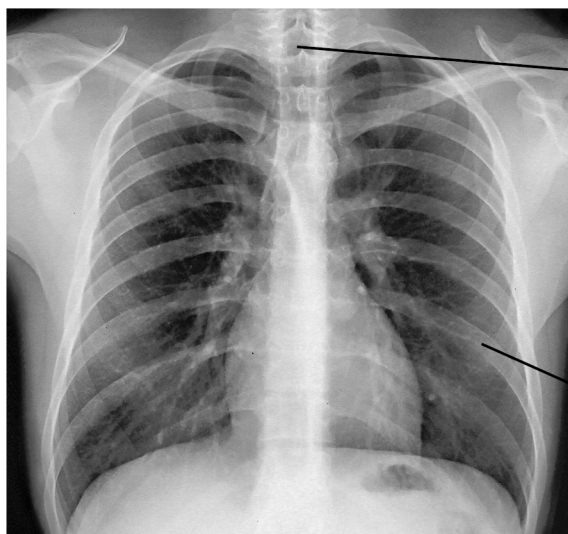
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0 6

Figure 10 is an X-ray of the human chest showing the human breathing system.

Figure 10



A _____

B _____

0 6 . 1

Label structures **A** and **B** in **Figure 10**.

[2 marks]

0 6 . 2

During breathing, oxygen moves from the air into the blood in the lungs.

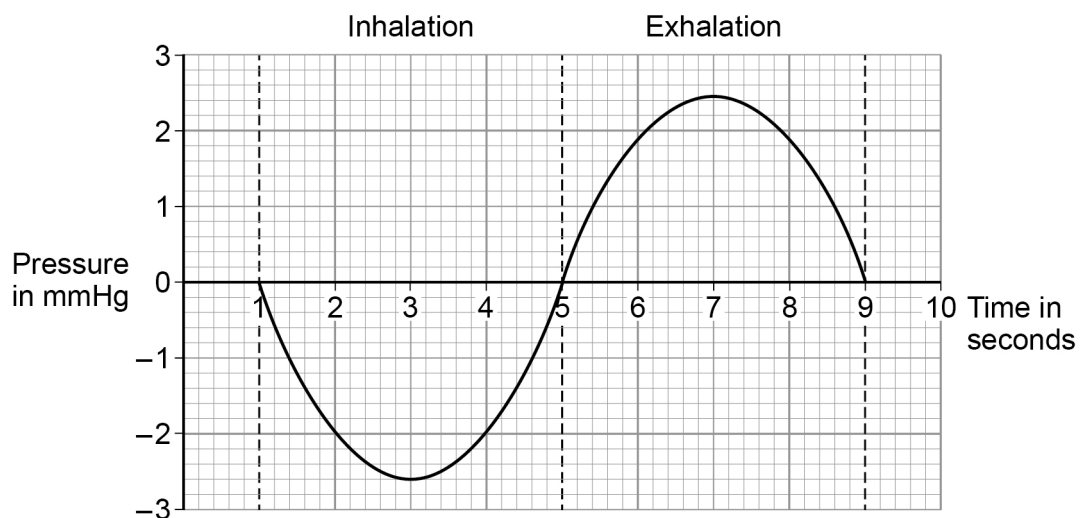
Explain how oxygen moves from the air into the blood.

[2 marks]



Figure 11 shows how pressure changes in the lungs during breathing.

Figure 11



0 6 . 3

Calculate the percentage increase in pressure between maximum inhalation and maximum exhalation.

Give your answer to three significant figures.

[3 marks]

Percentage increase = _____ %

Question 6 continues on the next page

Turn over ►



0 6 . 4

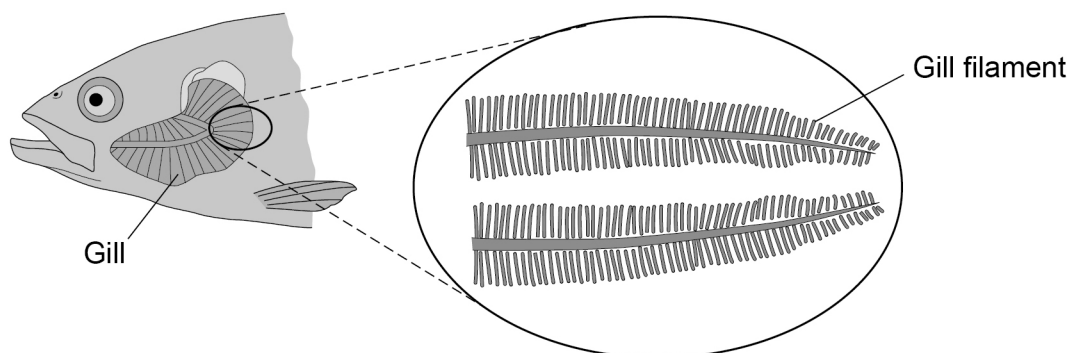
Explain what causes the pressure change during inhalation and how this causes air to enter the lungs.

[4 marks]

Fish have gills for gas exchange.

Figure 12 shows a fish head and gill.

Figure 12



0 6 . 5

Give **two** ways the gills are adapted to maximise gas exchange.

[2 marks]

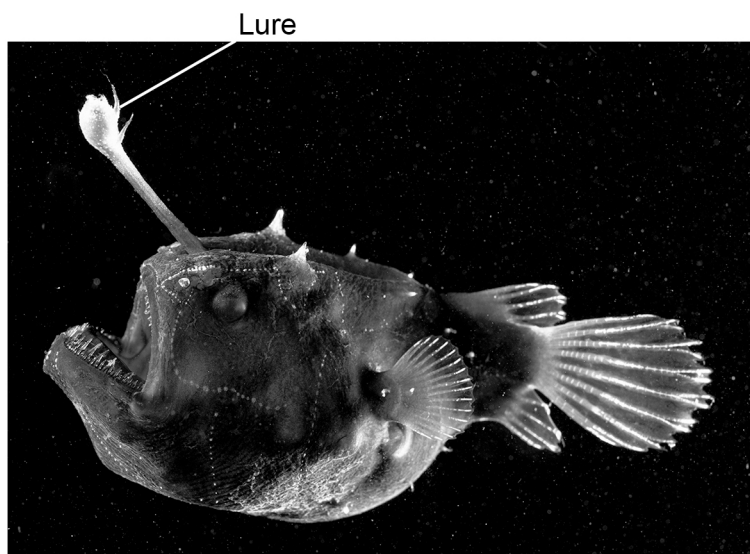
1 _____

2 _____



0 6 . 6 Figure 13 shows an angler fish.

Figure 13



Angler fish live at water depths of over 1000 m. Very few organisms can survive here.

At this depth there is very little light and the water temperature is near freezing point.

Angler fish have adapted to their environment in two ways:

1. They have a lure containing a high concentration of bioluminescent bacteria. Bioluminescent bacteria produce light.
2. They are very slow moving.

Explain how these adaptations help the angler fish to survive.

[4 marks]



0	7
---	---

Mosquitoes can transmit the malarial pathogen from one human to another.

0	7	.	1
---	---	---	---

What is a pathogen?

[1 mark]

0	7	.	2
---	---	---	---

A vaccine can be given to provide immunity against a pathogen.

Explain how a vaccine leads to immunity against a pathogen.

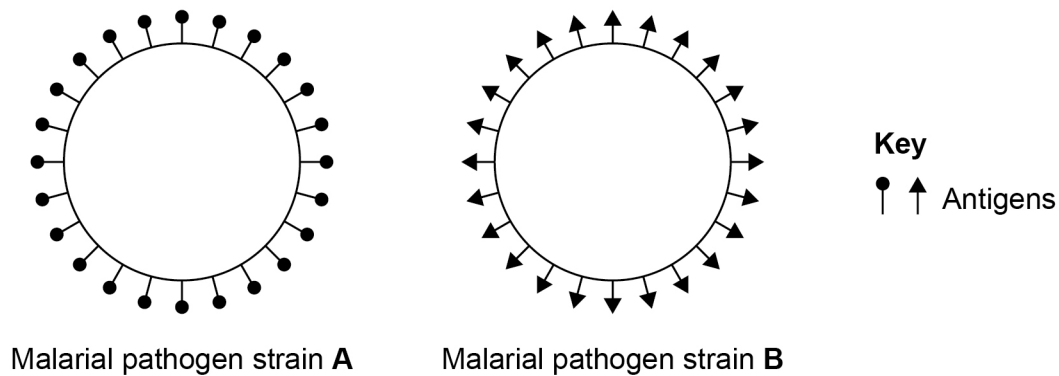
[3 marks]



A new strain of malarial pathogen **B** has evolved from a previous malarial pathogen **A**.

Figure 14 shows representations of the two strains.

Figure 14



0 7 . 3

Explain why people who already have immunity to malarial pathogen strain **A** may **not** be immune to strain **B**.

[3 marks]

Question 7 continues on the next page

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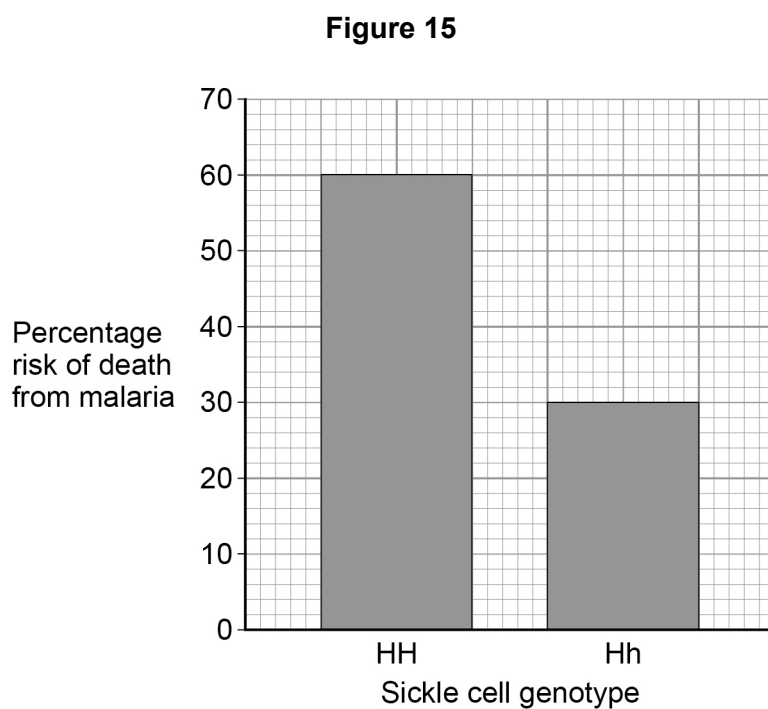
The malarial pathogen destroys red blood cells.

Sickle cell anaemia is an inherited disorder that affects haemoglobin.

The allele for normal haemoglobin is **H**. The allele that causes sickle cell is **h**.

In some parts of Africa where malaria occurs there is a high frequency of the genotype **Hh**.

Figure 15 shows how malaria affects people with each genotype differently.



0 7 4

Explain why there is a higher frequency of the genotype **Hh** where malaria occurs.

[3 marks]

END OF QUESTIONS



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Question number	Additional page, if required. Write the question numbers in the left-hand margin.



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